

TV GHOST CANCELLING BY LMS-RAT DIGITAL FILTER

Mohammad Reza Asharif*, Kazuo Murano**, and Mitsutoshi Hatori*

*Department of Electrical Engineering, Faculty of Engineering,
University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113, JAPAN

**Transmission Systems Laboratory, Fujitsu Laboratories Ltd.
1015 Kamikodanaka, Nakahara-ku, Kawasaki 211, JAPAN

I. INTRODUCTION

Recently several methods for automatic TV ghost cancelling are under active research. Generally speaking, these methods are categorized into two substantial groups: 1) TV ghost cancelling at RF stages including antenna method. 2) Cancellation at video stage.

In both of the mentioned above methods, there are many articles^[1-3]. As we know, there are two parameters for ghost evaluation i.e. the attenuation R and the delay τ . The ghost parameters are dependent on TV channel, location of the receiver antenna, and the weather condition which is a function of time and season. Therefore, the ghost parameters are variable and the method of cancellation should be capable of following these variations. Ghost may be produced by many multi-directional reflected signals, in this case, multi-ghosts appear on TV picture. It is difficult to treat multi-ghosts by the method of antenna or the method which is applied at RF stages^[2]. Therefore, the method of cancelling at video stage is thought to be more prospective, not only for its ability to process multi-ghosts or to construct a digital filter which is convenient for LSI, but also for its adaptive control nature. Therefore, this research is focused on the utilization of the adaptive digital filters for TV ghost cancelling. There are two types of digital filter:

The feedback and the feedforward types. The feedforward type generates the "grand-child" ghost if the filter does not have sufficient number of taps, while the feedback type does not have this kind of ghosts. In order to adjust the taps of the filter, the adaptive control technique is preferable for the simplicity of the control circuits. The Least Mean Square (LMS) algorithm is the original concept for the adaptation of the filter in the present work.

The joint optimizations of ghost and white noise is also considered. If one uses the inverse type of filter for TV ghost cancelling the input S/N will decrease that is the noise power will increase.

While, as we discuss later, by using the suitable test signal in the adaptive digital filter the joint optimizations of ghost and white noise will be obtained that is the noise power will not increase. Specially, the feedforward path in the digital filter is necessary for noise minimization^[4]. While the feedback path is needed to avoid the "grand-child" ghost.

In the realization of the adaptive recursive LMS digital filter for TV ghost cancellation the error samples have delay with respect to instantaneous input signal samples therefore the system may be over compensated. Here we have proposed a new method in order to avoid over compensation. This method is based on Random Access Tapping (RAT) control. It will be shown that the RAT method also prevents the system to enter into any pseudo optimum condition. In order to guarantee the stability of the RAT control we define a new function for signature operation in the LMS algorithm. The computer simulation results also confirm the capability of the recursive LMS algorithm and its improvement by RAT method for TV ghost cancellation.

The hardware design of the adaptive recursive LMS-RAT digital filter for TV ghost cancellation is described in detail. At the end the experimental results are given.

[5, 6, 7, 8]

II ADAPTIVE RECURSIVE LMS FILTER

Assuming the tapped delay line (TDL) filter with both the feedforward and the feedback paths as is shown in Fig. 1. If A and B are the vertical vectors of non-recursive and recursive tap coefficients, respectively and X , Y are column vector of the input, output signal samples series in the filter then we have:

$$A(n) = [a_0(n) a_1(n) a_2(n) \dots a_{NF}(n)]^T (1-a)$$

$$B(n) = [b_1(n) b_2(n) b_3(n) \dots b_{NB}(n)]^T (1-b)$$

$$X(n) = [x(n) \ x(n-1) \ \dots \ x(n-NF)]^T \quad (2-a)$$

$$Y(n) = [y(n-1) \ y(n-2) \ \dots \ y(n-NB)]^T \quad (2-b)$$

The output $y(n)$ is found by using discrete convolution:

$$Y(n) = \sum_{i=0}^{NF} a_i(n) \cdot x(n-i) + \sum_{j=1}^{NB} b_j(n) \cdot y(n-j) \quad (3)$$

Where NF and NB are the number of taps of the FF and FB paths, respectively. Equation (3) can be rewritten in the vector form as follows:

$$y(n) = A^T(n) \cdot X(n) + B^T(n) \cdot Y(n) \quad (4)$$

If the desired signal (without ghost or distortion) is $d(n)$ then the Error is:

$$\varepsilon(n) = y(n) - d(n) \quad (5)$$

then the MSE is:

$$\xi = E[\varepsilon^2(n)] \quad (6)$$

Where $E[\cdot]$ shows the expected value.

In the LMS algorithm the tap coefficients change proportional to the gradient of MSE and in the opposite direction, that is:

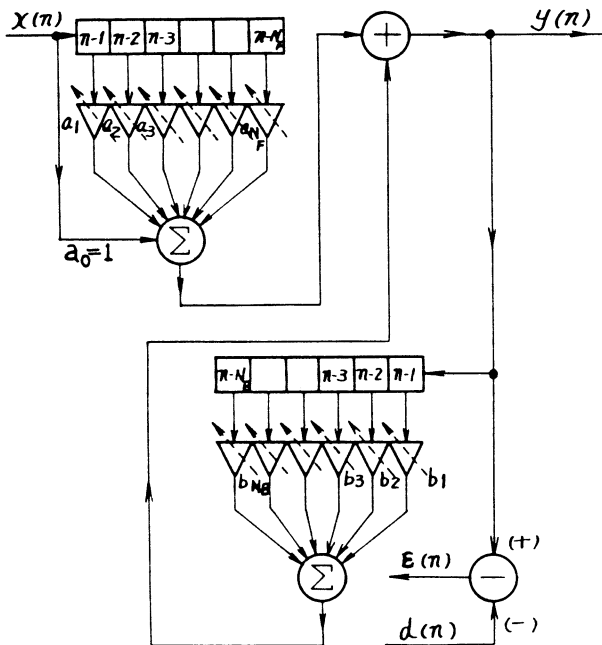


Fig. 1 Digital filter with FF & FB paths.

$$A(n+1) = A(n) - K_1 \cdot \nabla_{A(n)} E[\varepsilon^2(n)] \quad (7-a)$$

$$B(n+1) = B(n) - K_2 \cdot \nabla_{B(n)} E[\varepsilon^2(n)] \quad (7-b)$$

Suppose that $\nabla \varepsilon^2(n)$ is an unbiased estimation of $\nabla E[\varepsilon^2(n)]$

$$\text{That is : } E[\nabla \varepsilon^2(n)] = \nabla E[\varepsilon^2(n)] \quad (8)$$

Then the LMS algorithm becomes:

$$A(n+1) = A(n) - K_1 \cdot \nabla_{A(n)} [\varepsilon^2(n)] \quad (9-a)$$

$$B(n+1) = B(n) - K_2 \cdot \nabla_{B(n)} [\varepsilon^2(n)] \quad (9-b)$$

Substituting (4) into (5):

$$\varepsilon(n) = A^T(n) \cdot X(n) + B^T(n) \cdot Y(n) - d(n) \quad (10)$$

Then the gradients estimations are:

$$\nabla_{A(n)} [\varepsilon^2(n)] = 2 \varepsilon(n) \cdot X(n) \quad (11-a)$$

$$\nabla_{B(n)} [\varepsilon^2(n)] = 2 \varepsilon(n) \cdot Y(n) \quad (11-b)$$

And the LMS algorithm is:

$$A(n+1) = A(n) - 2K_1 \cdot \varepsilon(n) \cdot X(n) \quad (12)$$

$$B(n+1) = B(n) - 2K_2 \cdot \varepsilon(n) \cdot Y(n) \quad (13)$$

In Fig. 2 the realization of the recursive LMS algorithm (only FB path Eq. 13) is shown.

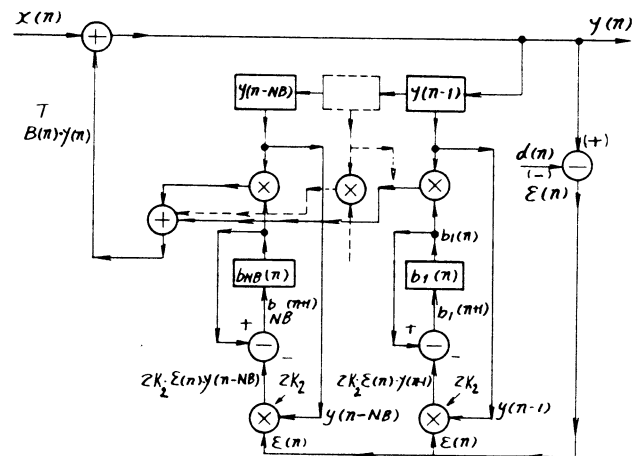


Fig. 2 Realization of the recursive LMS algorithm

III. TV GHOST CANCELLATION IN THE PRESENCE OF WHITE NOISE

III.1 TEST SIGNAL

One of the merit of the LMS algorithm is its performance in the presence of ghost and white noise. The frequency spectrum of the test signal has influence on the performance of the adaptive digital filter. If we use the unit pulse as the test signal because of its wide frequency spectrum the trained adaptive filter by such test signal has also wide frequency spectrum and the input white noise is not rejected by such filter. Whereas, if the frequency spectrum of the test signal is more similar to the one of the real video signal some part of noise spectrum will be cut out and consequently, the power of noise decreases. Considering also the technical problems for realization, the desired test signal is the one with two characteristics as follows:

- 1) Its power spectrum should be similar to the video power spectrum in order to achieve the joint minimization of ghost and noise.
- 2) The test signal should have a fixed phase and stable form in order to be memorized easily in memory cells as the reference signal.

For such test signal, we propose the horizontal synchronization pulse and the color burst in the blanking level of the color burst zero phase line from 11H to 16H in the vertical blanking interval. Therefore, there are six (or three) test signals in one frame (or one field) of TV signal. The spectrum of this test signal is more similar to the video spectrum than the one of the unit delta function. At the same time the phase of color burst in this test signal is always fixed which is easy to be memorized as the reference signal. In Fig. 3 the proposed test signal is shown. The power spectrum of this signal is given by the following equation:

$$\begin{aligned}
 F(w)^2 &= \left(T \frac{\sin wT/2}{wT/2} \right)^2 \\
 &+ \left(\frac{w_{sc}}{w_{sc}^2 - w^2} \sin 8\pi w/w_{sc} \right)^2 \\
 &+ 2 \left(T \frac{\sin wT/2}{wT/2} \right) \\
 &\left(\frac{w_{sc}}{w_{sc}^2 - w^2} \sin 8\pi w/w_{sc} \right) \sin wT_1 \quad (14)
 \end{aligned}$$

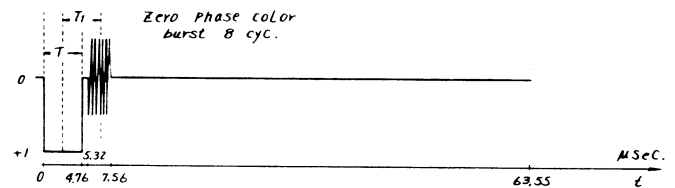


Fig. 3 Each of 11-16H (zero phase) used as test signal

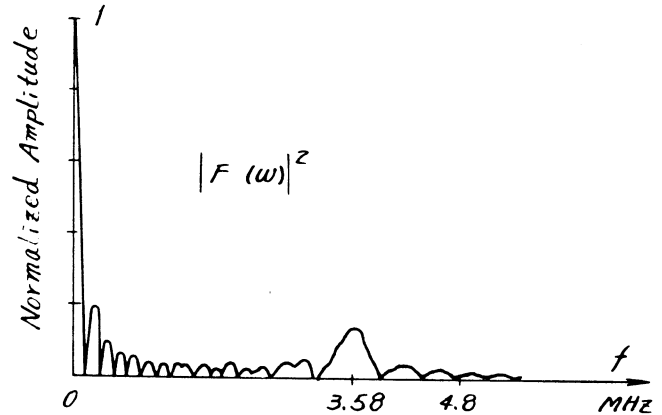


Fig. 4 Power spectrum of sync. pulse and color burst

In Fig. 4 the power spectrum of this test signal ($|F(w)|^2$) is shown.

Aside of the mentioned specifications for this test signal, there is no interference from the previous line to the test signal. Because each line before test signal has long period of blanking level. If the ghost of the previous line appears inside of the test signal, the adaptation algorithm confuses in detecting of the ghost delay and it leads to over compensate the tap coefficient which does not have direct correlation with the real ghost delay.

The maximum possible ghost delay for the proposed test signal without interfering to the next line is 600 samples (each sample has $1/3f_{sc}$ rate) which is $\frac{600}{680} \times 100 = 88\%$ of

horizontal line and $\frac{600}{630} \times 100 = 95\%$ of the visible width of the TV picture (CRT width).

The line 17 H is another candidate as the test signal. Of course the spectrum of this signal is more similar to the video spectrum than each of the lines from 11 H to 16 H. However, line 17 H contains the $\sin 20T$ pulse which is modulating the color sub-carrier with running phase. Therefore, using 17 H as the test signal requires a technically complicated phase adaptation control process.

III.2 COMPUTER SIMULATION ASSUMING GHOST AND WHITE NOISE

In order to simulate the adaptive digital filter it is necessary to take samples from the proposed test signal shown in Fig. 3. The sampling frequency is $3f_{sc}$ (where $f_{sc} = 3.58$ MHz is color burst frequency). Therefore, the sampling rate is $\tau = \frac{1}{3f_{sc}} = 93$ nsec. In Fig. 5 the sampled test signal is shown. In the computer simulation we have truncated 600 zero samples of black level into 80 samples. Therefore, 160 samples are representing one test signal. During each test signal 160 iterations for adjusting of the tap coefficients are performed (one iteration per sample). The taps will be frozen after the 160th sample until the first sample of the next test signal appears. Here, the digital filter with both feedforward and feedback paths is considered, Fig. 1. The recursive LMS algorithm (Eqs. (12), (13)) is employed for taps adaptation of the filter. The scalar form of this algorithm is:

$$a_i(n+1) = a_i(n) - 2\Delta_{FF} \cdot \epsilon(n) \cdot x(n-i) \quad (15-a)$$

$$b_j(n+1) = b_j(n) - 2\Delta_{FB} \cdot \epsilon(n) \cdot y(n-j) \quad (15-b)$$

Suppose that the input $x(n)$ is degraded by ghost signal and white noise with the following patterns:

$$x(n) = d(n) + 0.1d(n-2) + 0.1d(n-3) + w(n) \quad (16)$$

Where $d(n)$ is the desired (test) signal. The noise signal $w(n)$ is uncorrelated white noise with power σ^2 :

$$E[w(n) \cdot w(m)] = \sigma^2 \cdot \delta(m-n) \quad (17)$$

The input S/N is defined as follows:

$$\frac{S}{N} = -10 \log \sigma^2 = -20 \log \sigma \text{ dB} \quad (18)$$

Several amounts of input S/N from 10 dB to 50dB are examined in order to evaluate the performance of the LMS algorithm for adaptation of the digital filter in the presence of noise. The output D/U is defined by the following equation:

$$D/U = -10 \log \left(\sum_{i=1}^{160} \epsilon^2(i)/160 \right) \text{ dB} \quad (19)$$

As the results of the simulation in all noisy cases, the output D/U approaches to the input S/N after convergence of the adaptation process. In Fig. 6-a the output D/U versus number of iteration when input $\frac{S}{N} = 45$ dB is shown. The process of tap coefficients adaptation is shown in Fig. 6-b. Here, we have used 10 taps for feedforward path and 10 taps for feedback path. The step sizes $\Delta_{FF} = \Delta_{FB} = 0.01$ are chosen. The process of adaptation reaches to acceptable state, that is D/U converges to input S/N during 5759 iterations. Remembering that there are 160 iterations in one test signal so that $\frac{5759}{160} = 36$ test signals are required. Having 6 test signals in each TV frame, we need $\frac{36}{6} = 6$ frames or $\frac{6}{30} = 0.2$ sec. for adjusting the digital filter using recursive LMS algorithm.

As we have seen the output D/U is approaching to input S/N when we use the LMS algorithm. It is easy to verify that by using inverse type filter the noise power will increase i.e. S/N decreases. In the present simulation for LMS algorithm we suppose that $a_0 = 1$ and others tap coefficients are adaptive control. It is shown that^[4] by the adaptation of the through tap, i.e. a_0 , according to LMS algorithm as well as others taps, the output D/U is obviously increasing from input S/N that is noise power reduces. Of course, in this case the low pass characteristics of the filter is reducing the sharpness of the signal, thus the picture becomes blurred.

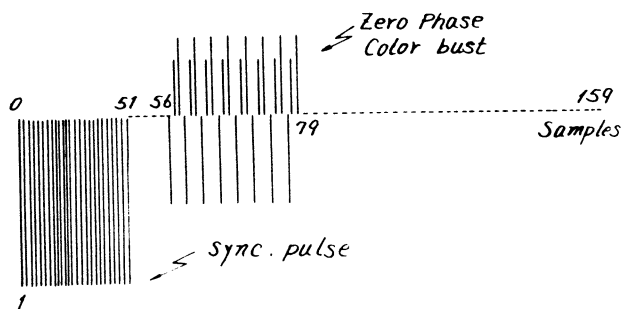


Fig. 5 Sampled test signal by 3 fsc.

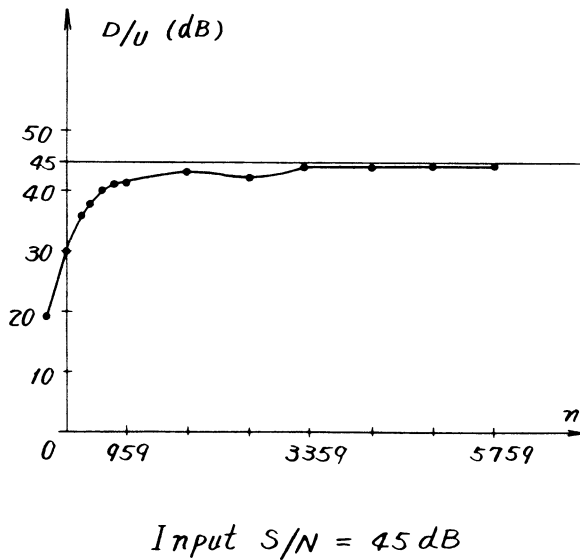


Fig. 6-a D/U versus iteration number
for $\frac{S}{N} = 45\text{ dB}$

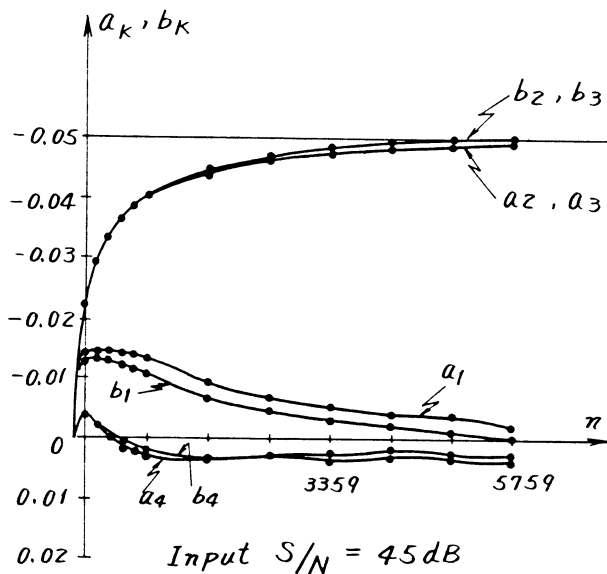


Fig. 6-b Taps adaptation curves for $\frac{S}{N} = 45\text{ dB}$

IV.1 RANDOM ACCESS TAPPING (RAT) CONTROL OF THE ADAPTIVE LMS FILTERS

Here, we shall introduce a new method for tap adjustment which is based on selecting some of the taps at random for adaptation at each iteration. The purpose of this method is to avoid over-compensation or pseudo-optimum conditions which are the problems due to undesirable delay in the implemented model and simplification of algorithm for realization. The method of Random Access Tapping (RAT) control gives good solution to such kinds of problems.

For simple realization of the LMS algorithm expressed by Eqs. (12), (13) we take signature of the compensation parts in the algorithm, that is:

$$a_i(n+1) = a_i(n) - \Delta \cdot \text{Sign}[\varepsilon(n)] \cdot$$

$$\text{Sign}[x(n-i)] \quad (20-a)$$

$$b_j(n+1) = b_j(n) - \Delta \cdot \text{Sign}[\varepsilon(n)] \cdot$$

$$\text{Sign}[y(n-j)] \quad (20-b)$$

We suppose that $x(n)$, $y(n)$ are quantized into 6 bits and $a_i(n)$, $b_j(n)$ into 11 bits. The algorithm in Eqs. (20-a), (20-b) can be easily realized by using up/down counter and a simple exclusive "OR" gate as is shown in Fig. 7. The sampling frequency of the video input signal is $3f_{sc} = 10.7\text{ MHz}$. As we have explained in part III the refreshment of the taps are performed once at each sample i.e. each $\frac{1}{10.7} = 93\text{ nsec}$. In such a high speed adaptive digital filter it is difficult to detect the error during one sampling period using currently available devices. Therefore, the algorithm is changed as follows:

$$a_i(n+1) = a_i(n) - \Delta \cdot \text{Sign}[\varepsilon(n-k)] \cdot$$

$$\text{Sign}[x(n-k-i)] \quad (21-a)$$

$$b_j(n+1) = b_j(n) - \Delta \cdot \text{Sign}[\varepsilon(n-k)] \cdot$$

$$\text{Sign}[y(n-k-j)] \quad (21-b)$$

where k is the required samples period for detection of one error sample.

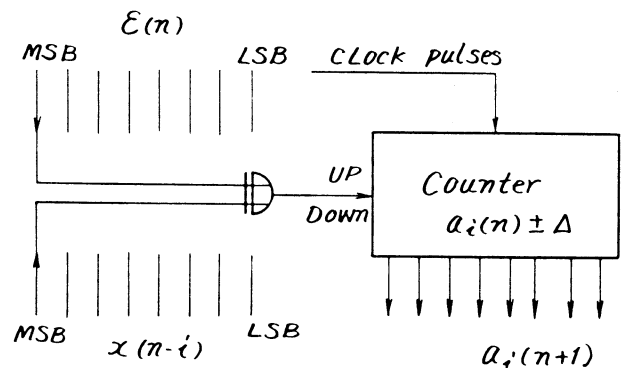


Fig. 7 Tap's adjustment process

However, this algorithm may over-compensate the digital filter, since during k samples the taps will be adapted by the same compensation order as error has been delayed by k samples, a look at Fig. 8 may describe this fact.

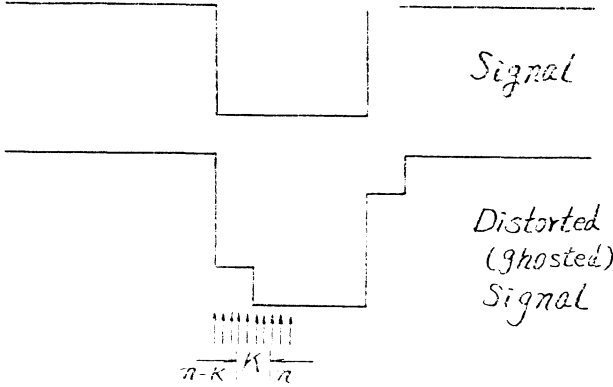


Fig. 8 Over-compensation due to lag of error

In order to avoid this over-compensation it seems that if the refreshments of the taps are performed at every k' samples ($k' \geq k$) then, the error samples could be obtained before refreshment and consequently, the over-compensation is prevented. However, as we shall explain below, there is possibility of pseudo-optimum condition. Because, the algorithm is written as follows:

$$a_i(n+k') = a_i(n) - \Delta \cdot \text{Sign}[\varepsilon(n+k'-k) \cdot x(n+k'-k-i)]$$

$$a_i(n+2k') = a_i(n+k') - \Delta \cdot \text{Sign}[\varepsilon(n+2k'-k) \cdot x(n+2k'-k-i)]$$

$$\cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot$$

$$a_i(n+N') = a_i(n+N'-k') - \Delta \cdot \text{Sign}[\varepsilon(n+N'-k) \cdot x(n+N'-k-i)]$$

$$a_i(n+N') = a_i(n) - \Delta \cdot \sum_{M=1}^{N'/k'} \text{Sign}[\varepsilon(n+M \cdot k'-k) \cdot x(n+M \cdot k'-k-i)] \quad (22)$$

In the right hand side of Eqs.(22) the summation of signatures terms may become zero for some N' where $N' = L \cdot N$ (L is an integer number and N is the number of samples in one test signal).

In this case, the tap is frozen because:

$$a_i(n+N') = a_i(n) \quad (23)$$

So that after N' samples the value of the tap is the same while the locations of samples n and $n+N'$ are the same in the test signal (because $N' = L \cdot N$). Therefore, continuing of this algorithm is meaningless and the adjustment of the tap enters into a repetition cycle and the process will be trapped by pseudo-optimum state.

The real optimum condition is when the gradient of MSE is zero, that is, according to Eq. (11-a) when:

$$\varepsilon(n) \cdot X(n-i) = 0 \quad (24)$$

From the past discussion we conclude that the separation of the refreshments by k' samples may mis-lead the system to a pseudo-optimum condition. Here, we propose a new method based on Random Access Tapping(RAT) in which the refreshment performed at each sample but for some of the taps chosen at random. If the probability of each tap for being adjusted is $1/k'$, then the average separation of the refreshment moments for each tap is k' . Therefore, the over-compensation is prevented and at the same time because of random nature of the process it is quite unlikely to have a pseudo-optimum condition. If the separation of the refreshment moments is k' (fixed not random) in order to avoid pseudo-optimum condition it is also advisable to use RAT method. In this case, the probability of each tap for being adjusted is not necessarily $1/k'$, but it could be with higher probability. After perturbing the system from pseudo equilibrium condition by RAT, the system has more chance to go towards the real optimum condition. However, keeping the system in the desired optimum condition needs some technique which we shall deal with later.

IV.2 COMPUTER SIMULATION FOR "RAT" CONTROL OF THE ADAPTIVE LMS FILTERS

In the present simulations we have employed the RAT control method in the digital filter in order to obtain a TV ghost canceller with good performance. The test signal is the same as shown in Fig. 3 and the sampled test signal by 3fsc is shown in Fig.5. Assuming a strong multi-ghosts pattern:

$$x(n) = d(n) + 0.1 \sum_{k=1}^9 d(n-k) \quad (25)$$

And suppose that the error samples are delayed by four samples therefore the algorithm is: ($\Delta=0.001$)

$$a_i(n+1) = a_i(n) - \Delta \cdot \text{Sign}[\varepsilon(n-4)] \cdot \text{Sign}[x(n-4-i)] \quad (26-a)$$

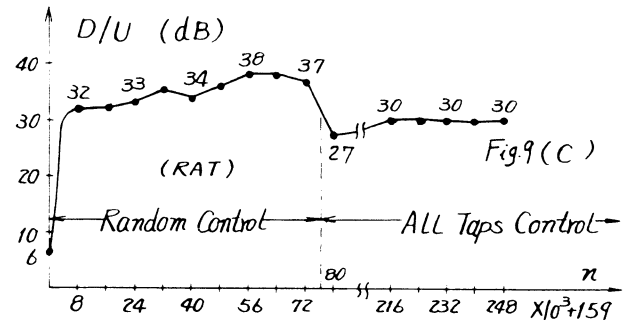
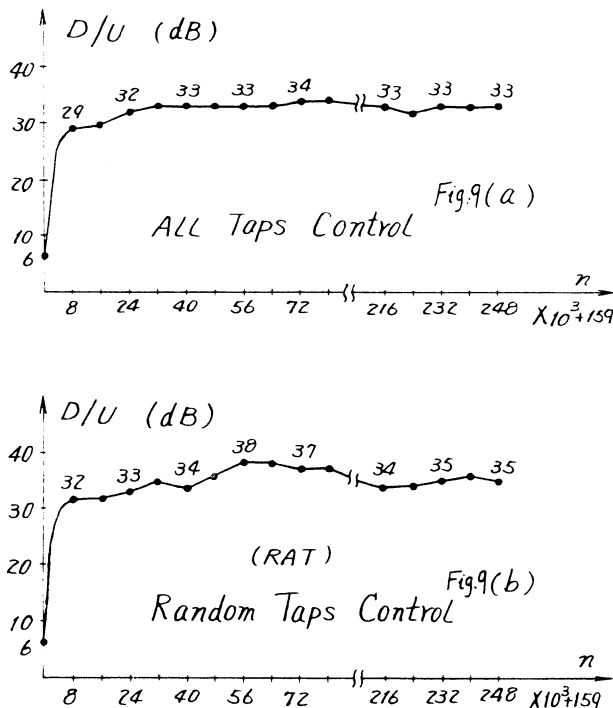
$$b_j(n+1) = b_j(n) - \Delta \cdot \text{Sign}[\varepsilon(n-4)] \cdot \text{Sign}[y(n-4-j)] \quad (26-b)$$

Using all taps in each adjustment the learning curve of D/U (Eq.(19)) with respect to iteration number is shown in Fig. 9-a. Since the error samples have delay this algorithm may be overcompensated. In order to avoid this problem we should utilize the RAT control. In Fig. 9-b the result of RAT control is shown, where at each iteration two taps out of ten taps are adjusted randomly ($P = \frac{1}{5}$) (the same taps for FF & FB).

Comparing Fig. 9-a with 9-b we observe that D/U is increasing when the RAT control is used. In order to prove the effectiveness of the RAT control, we stop the process of the RAT at 80000th sample (iteration) and at this moment all taps are participating in the adjustment. In this case, as it is shown in Fig. 9-c, the D/U is suddenly decreasing (to 27 dB) at the moment of stopping the RAT. While, continuing the RAT control would result 10~5 dB higher D/U.

Now, we try the following form of the algorithm where the refreshment is executed once at every five samples ($n = 5k$), and assuming the same pattern of ghost as in Eq. (25) ($\Delta=0.001$)

Fig. 9 Evaluation of the RAT control in the LMS digital filter with error lag



$$a_i(n+1) = a_i(n-4) - \Delta \cdot \text{Sign}[\varepsilon(n-3)] \cdot \text{Sign}[x(n-3-i)] \quad (27-a)$$

$$b_j(n+1) = b_j(n-4) - \Delta \cdot \text{Sign}[\varepsilon(n-3)] \cdot \text{Sign}[y(n-3-j)] \quad (27-b)$$

As it is shown in Fig. 10-a the system goes into an equilibrium condition with D/U=15 dB which is actually a pseudo-optimum state. After using the RAT method for the same algorithm the system will come out from this pseudo-optimum state and D/U will increase more than 10 dB. Here, five taps are chosen at random for being adjusted at each iteration, the result is shown in Fig. 10-b.

Since the RAT control uses the random numbers in order to select the taps for adaptation, there is the possibility to have unstable values of D/U. That is, D/U may decrease from a desirable value to a lower one (see Fig. 10-b). This is because the RAT control may select some taps which are not sufficiently orthogonal to each others. If one traps the process of the RAT at the highest value of D/U the stability of the RAT method will be guaranteed.

In order to avoid any unstable state we shall define a new signature operator, in the LMS algorithm, with the three levels -1, 0, +1, shown in Fig. 11, in contrary with two levels of the normal signature operator.

In this case while the error sample is inside the zero area of the signature the process of adaptation is stopped. When the D/U is high enough all error samples, in one period of the test signal, are lying inside the zero area and filter's taps will be frozen so that D/U remains at the desirable high level. The operator of signature which was defined in Fig. 11 is called "Lazy or Dull" signature. By this method we actually "catch" the error at low level (or D/U at high level) which is found by the RAT control.

Then we say "The "RAT" is caught by the "cage" of the lazy signature". The computer simulation result supports this compensation method.

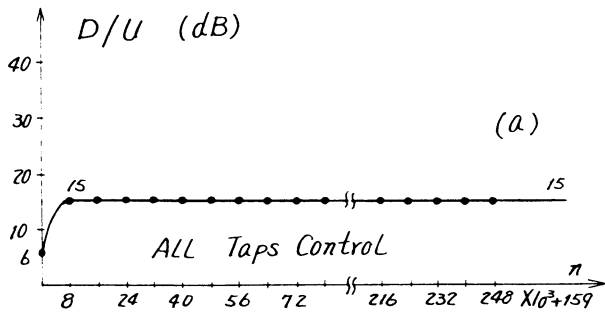


Fig. 10-a Pseudo-optimum state occurs in the LMS digital filter ($D/U = 15\text{dB}$).

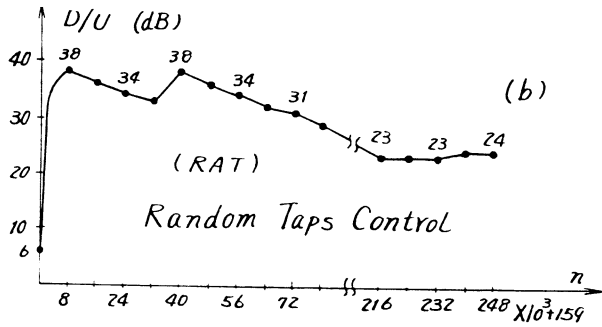


Fig. 10-b RAT control perturbs the system from pseudo-optimum state

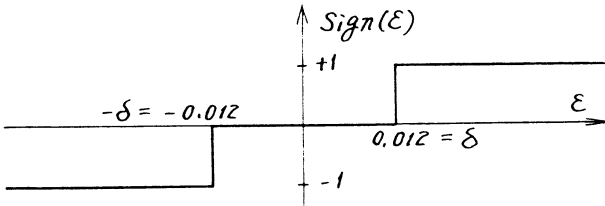


Fig. 11 Three levels (lazy) signature operator

IV.3 COMPUTER SIMULATION FOR COMPENSATING THE RAT CONTROL BY LAZY SIGN. OPERATOR

Assume the same test signal in Fig. 3 and a double ghosts pattern as follows:

$$x(n) = d(n) + 0.1d(n-2) + 0.1d(n-3) \quad (28)$$

with initial value of $D/U = 18.8$ dB. We employ the following form of the algorithm where error samples have been delayed by three samples:

$$a_i(n+1) = a_i(n) - \Delta \cdot \text{Sign}[\epsilon(n-3)] \cdot \text{Sign}[x(n-3-i)] \quad (29-a)$$

$$b_j(n+1) = b_j(n) - \Delta \cdot \text{Sign}[\epsilon(n-3)] \cdot \text{Sign}[y(n-3-j)] \quad (29-b)$$

The step size $\Delta = 0.001$ and the width of zero area in Fig. 11 is chosen $\delta = 0.012$. The method of RAT control is also applied, each time refreshing two taps out of ten taps at random (for both FF & FB) by using uniformly distributed random number generator. The output D/U , which is computed by Eq.(19), is plotted in Fig. 12. Here, the input and output are quantized into six bits therefore the quantization noise is:

$$10 \log (2^{2n} \times 12) = 46 \text{ dB} \quad (30)$$

for: $n = 6$

In order to certify the effectiveness of RAT control, let us consider the case of Fig. 12 with the same pattern of ghost and midtread quantization but allowing the taps to be refreshed all together. The result is depicted in Fig. 13. As we see the stability of the D/U is not met such as the RAT control would provide in Fig. 12. The reason is that, when we refresh all taps together at each sample the over-compensation may occur for the algorithm since the error is delayed by three samples (see Eqs.(29-a), (29-b)). As we have explained in part IV.1 the RAT control protects the algorithm from over-compensation.

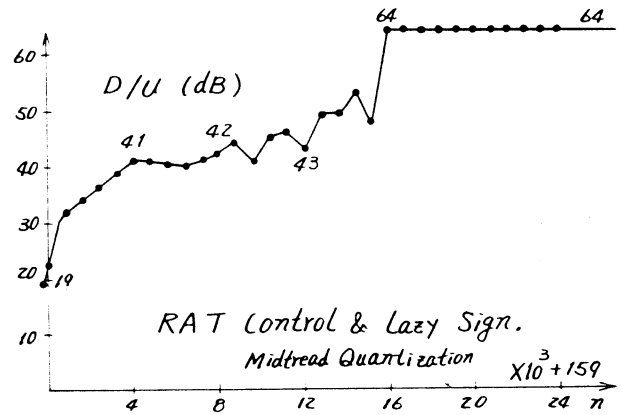


Fig. 12 Stabilization of the RAT control by lazy signature operator in the LMS digital filter

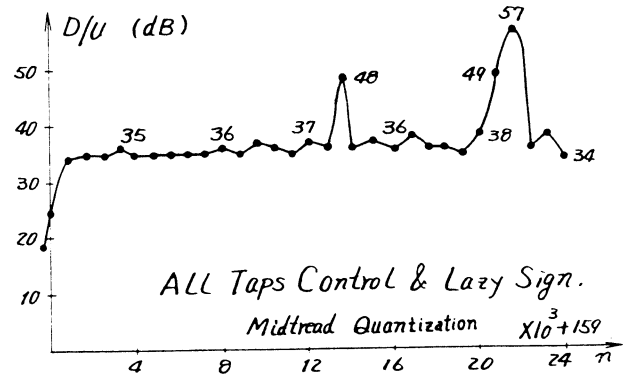


Fig. 13 All taps control results in over-compensation due to error lag

In general, we conclude that in order to improve the performance and stability of the easily implemented and high speed adaptive recursive LMS digital filter, both the RAT control and the lazy signature operator are necessary.

In the next part we shall explain the design of the TV ghost canceller by using adaptive recursive LMS digital filter.

V. DESIGN AND IMPLEMENTATION OF THE ADAPTIVE RECURSIVE LMS DIGITAL FILTER FOR TV GHOST CANCELLING

V.1 INTRODUCTION

In this part we shall explain the realization of the system. The experimental system objectives are:

- 1) Confirm the feasibility of the model regarding hardware realization.
- 2) Evaluate the ghost cancelling capability of the proposed algorithm using real pictures.

The used technology in the present work is based on TTL off-the-shelf devices. The motivation for this choice is to obtain reasonable size and power consumption. Mostly Schotcky TTL are used.

V.2 HARDWARE

The system of TV ghost canceller consists of three parts:

- (1) Interfacing circuits for connecting the digital filter to input-output video stages: A/D and D/A convertors and clamper circuit.
- (2) Control circuits which generate the necessary control pulses for manipulating the filter. The following circuits are used in the control section:
 - (a) Clock generator and phase-locked loop (PLL) circuit.
 - (b) Horizontal and vertical related sync. circuit.
 - (c) Reference signal generator and error detector circuit.
 - (d) Random Access Tapping (RAT) control and manual adjusting circuit.
- (3) Digital filter which is the normal transversal filter having both the feedforward and feedback paths.

In Fig. 14 the block diagram of the TV ghost canceller is shown.

The test signal, as we have explained in part IV, is each of three lines of 11 H ~ 16 H with zero phase color burst, Fig. 3.

Therefore, the taps should be refreshed only during this test signal. In other words, the test signal duration should be detected and during this time we need some control pulses to allow taps adjustment. The taps adjustment are performed for each sample of the test signal. In the LMS algorithm, as we have seen before Eq. (21-a), (21-b), the error samples are required to be detected. Using Read Only Memory (ROM) chips for memorization of the reference signal, it is possible to obtain the error samples. The ROM's address is decided by counters with sampling frequency clock i.e. 3 fsc. The sync. pulse at the beginning of each test signal ($2T_H$ period) is detected by magnitude comparator then clear pulses are made for resetting the counter at this time. The following relation is valid for sampling period in the NTSC system:

$$2T_H = 1365T_S$$

Using the test signal as described in Fig. 3, the difference between the reference signal stored in the ROM and the actual input signal may become quite large at the rising and falling edges of the horizontal sync. pulse. Slight variation in the slope angle would cause large errors in amplitude.

Similar situation occurs at the beginning and end of the color bursts. Therefore, in order to reduce misadjustments due to the ambiguity moments of the reference signal, it is advisable to stop the adaptation process at these moments.

As we have explained there are control pulses which allow the adaptation of the taps during the test signal. If these control pulses have prohibition moments according to the ambiguity periods in the test signal then the filter is saved from any misadjustment. In this case a simple window shape of control pulse is replaced by three windows for each test signal which we call it Adaptation Enable Pulse (AEP) in Fig. 15-a the time chart for various control pulses are shown.

Since there are three test signals in the vertical blanking interval of each TV field we need three AEP control pulses per field. These pulses are also written in a ROM chip whose address is decided by a counter with f_H (line frequency) as clock.

Since the address should be the same at the beginning of each field the counter will be cleared when the equalization pulses at the beginning of each field arrive.

The rate of equalization pulses is $2f_H$ in contrast with the sync. pulses which have f_H rate, therefore, it is possible to detect the moment of equalization pulses or Vertical Blanking Interval (VBI) at each field and consequently clear the counter just before the test signal arrival (see time chart in Fig. 15-b).

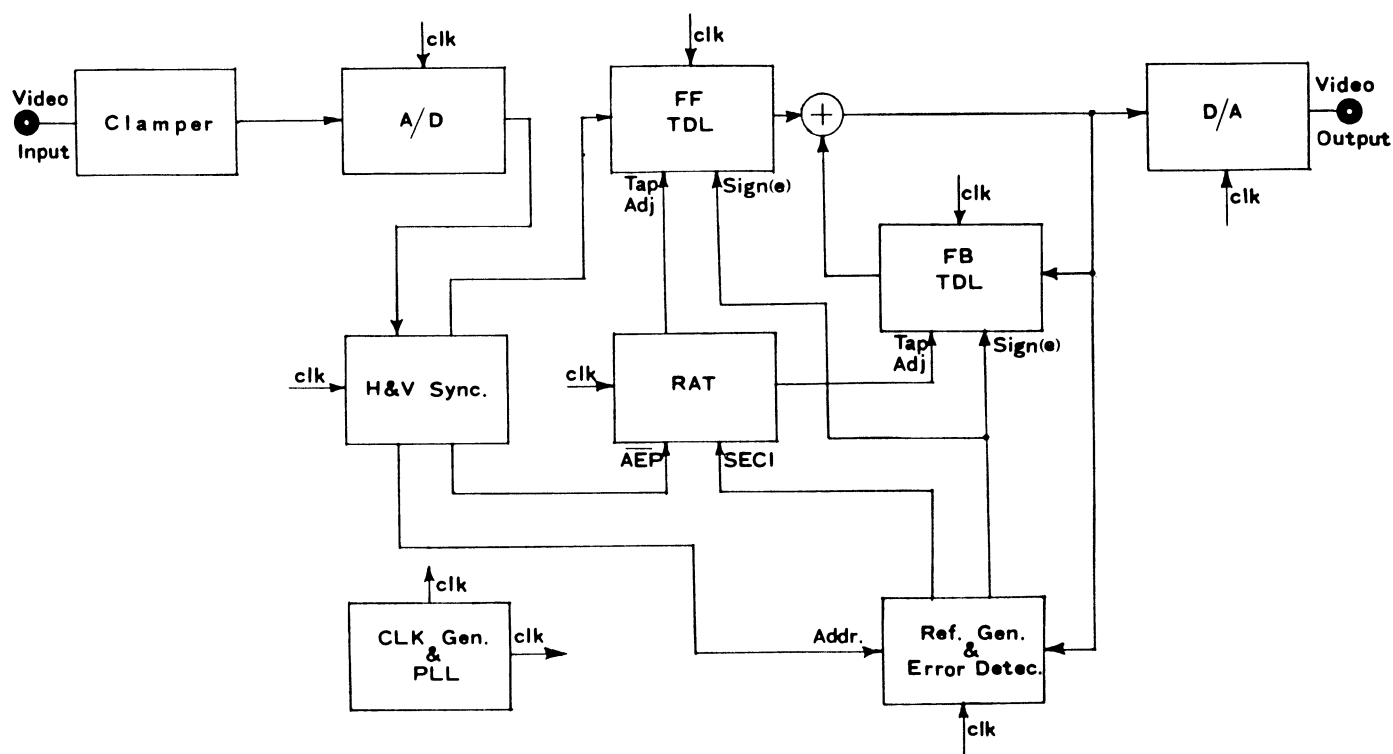


Fig. 14-a Block diagram of TV ghost canceller

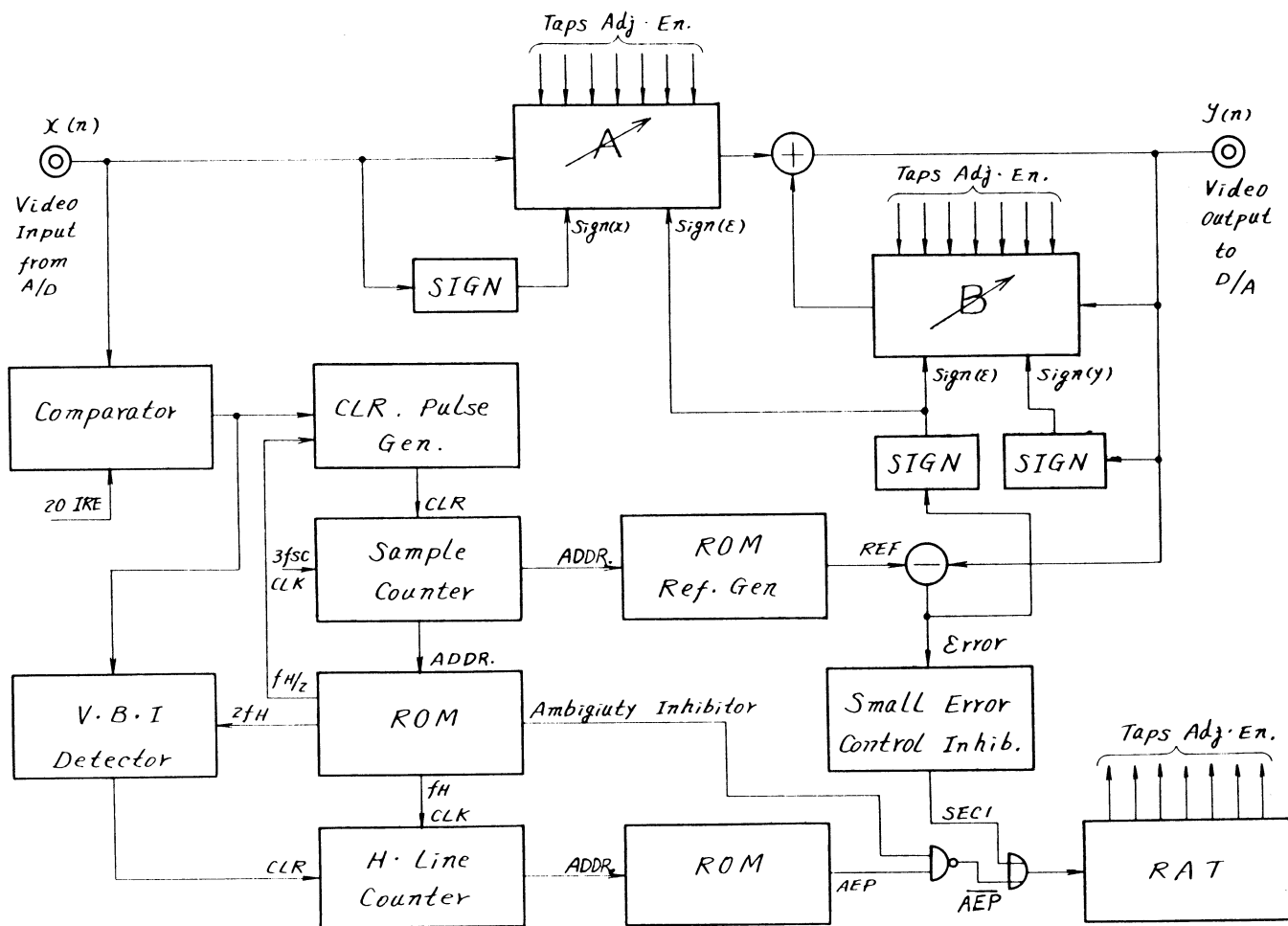


Fig. 14-b Detail block diagram of TV ghost
canceller

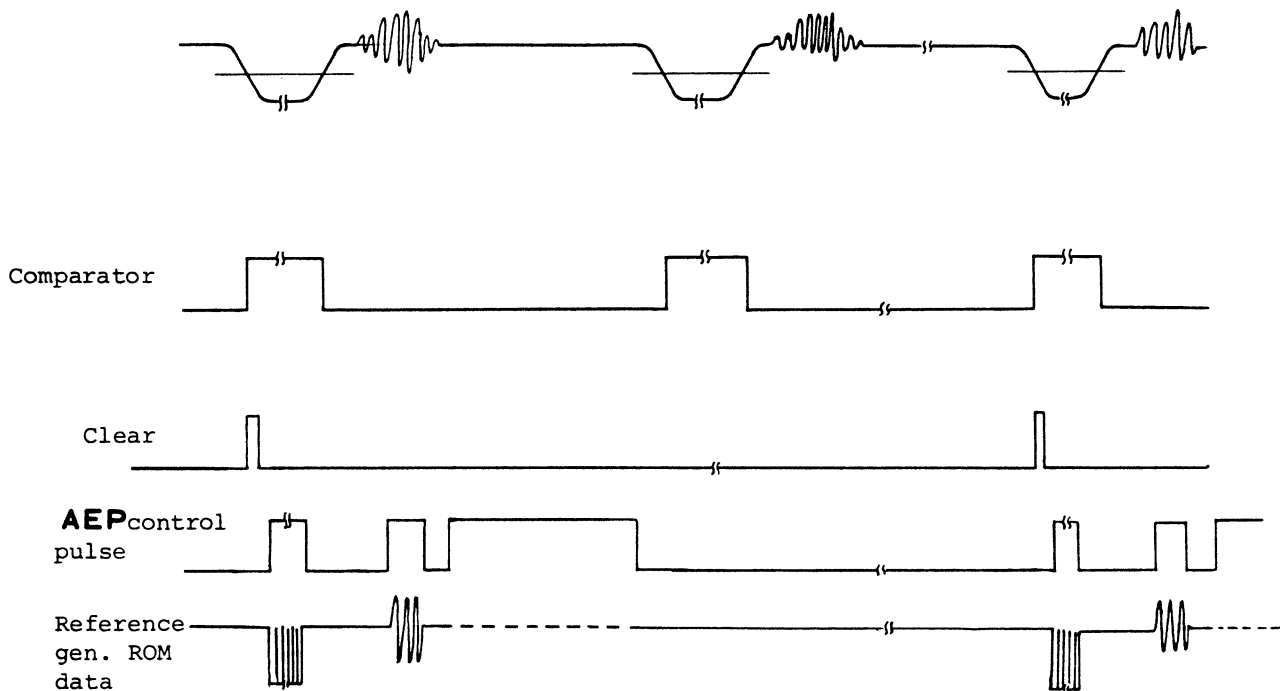


Fig. 15-a Time chart

In the clock generator and PLL circuit a signal with $3f_{sc}$ frequency is generated. The phase of this signal is locked to the received color burst with f_{sc} frequency by PLL circuit. Then the clock pulses for digital circuits are generated. The clock pulses have period of 93 nsec. ($\frac{1}{3f_{sc}}$). In order to realize the weighted samples in the digital filter the multiplier clocks should be provided. Since the input output clocks of the multiplier should be separated as long as 60 nsec. using TTL devices we should make another subclock pulses from the original clock pulse. If ϕ_1 is the original clock phase, clocks with ϕ_2 and ϕ_3 phases will be made by delaying the original clock pulse as it is shown in Fig. 16. Using ϕ_2 and ϕ_3 clocks as input and output clocks of the multiplier the required condition will be met and at the same time the computation of a weighted sample is performed in the minimum number of clock pulses. Therefore, the error samples have the shortest delay time (as we have explained in part IV.1 the error has unavoidable delay which causes over-compensation in the algorithm).

In the digital filter the two's complement system is used for coding the digital levels of the signal and the values of the tap coefficients. As we know in this system of coding the most significant bit (MSB) is representing the signature of the value. In the LMS algorithm of Eqs. (21-a), (21-b), the compensation parts include the sign of error and input and output signals.

In case of very small error signal such as 00000001 and, for instance, input 01111111 the compensator causes one step size (Δ) increment in the tap coefficient because the sign bit is plus (0). Therefore, as we recognize, the compensator is either plus (0) or minus (1) in any case. This fact causes the taps to be always in variation even near the optimum values and the system stability degrades. In order to avoid this problem we have made a zero area for signature as it was described in part IV.3. In this case for the value of error which lies inside the zero area there is no compensation for the taps. The zero area is chosen correspond to the area with first 5 bits to be all zero or all one. That is, the error is assumed to be zero for values between 00000¹¹¹xxx and 11111⁰⁰⁰xxx. Therefore, we need another control pulse for performing this job which we shall call Small Error Control Inhibitor (SECI) pulse. The SECI pulses are made in error detector board circuit.

Now, the adaptation of the taps is controlled by AEP and SECI pulses. Whenever AEP + SECI has low logic level (0) the adjustment of taps are allowed otherwise the taps are frozen. In order to realize the method of Random Access Tapping (RAT) control, which was explained in part IV.1, we need one control signal for each tap.

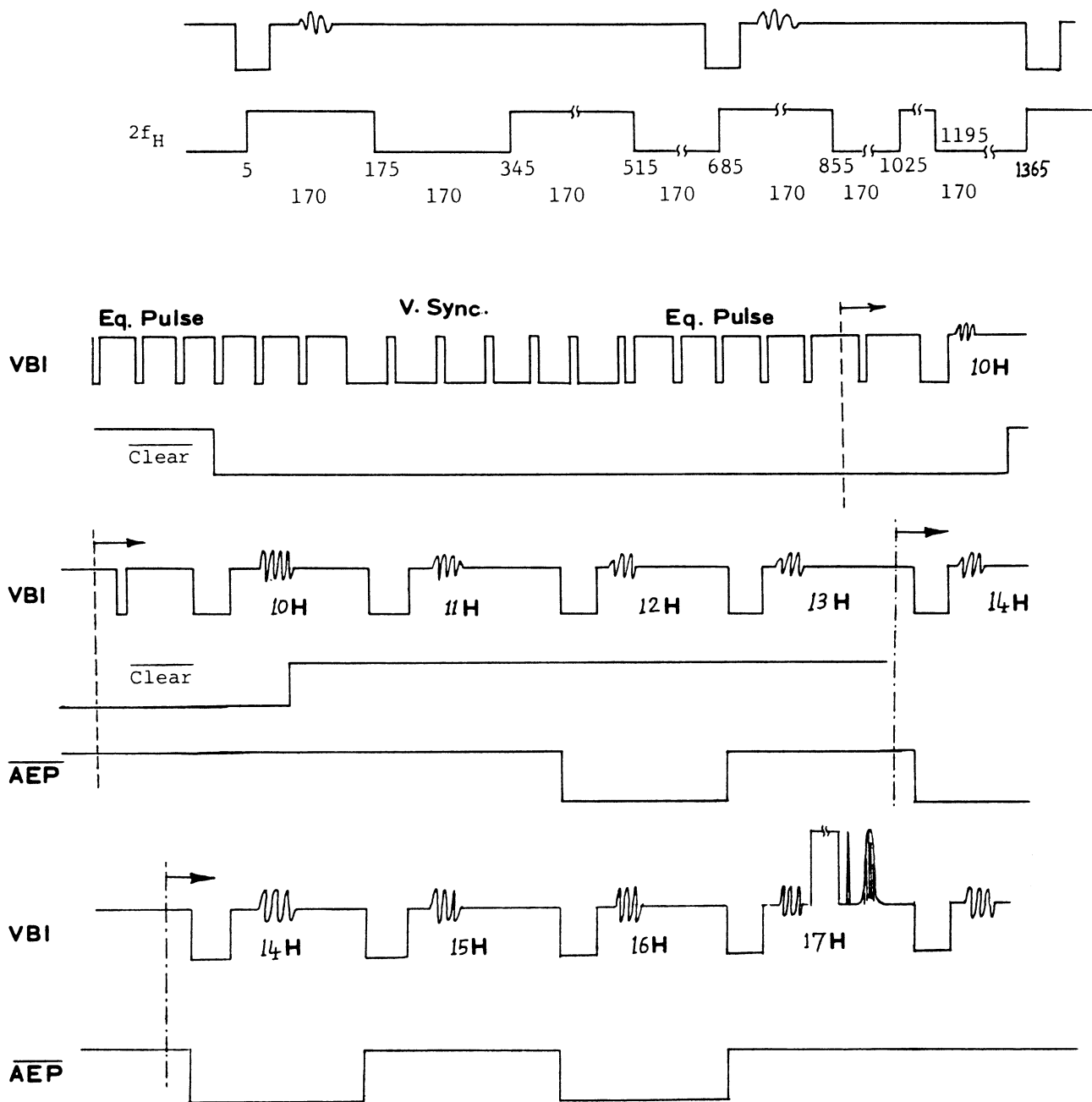


Fig. 15-b Time chart for detection of 3 test signals by AEP (without considering ambiguity moments prohibition)

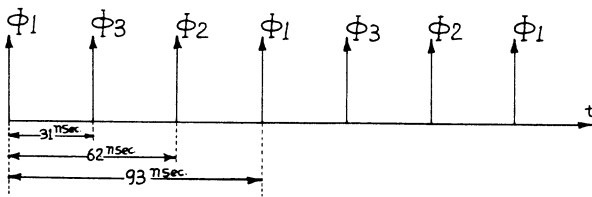


Fig. 16 Time chart of clocks

For this purpose we have used ROM chips to generate random sequence data. The addresses of the ROMs are provided by a free running counter. For each sample (address) there are 16 bits of random data which select some of the taps (there are 16 taps; 6 taps FF and 10 taps FB) at random for being adjusted. Of course the refreshment of the selected taps are performed only when both control pulses AEP and SECI have low logic levels.

Each pair of taps, and related circuits for obtaining the weighted samples, are designed in a single board. For each tap there is one adjust pulse coming from RAT circuit. When the random sequence chooses one tap for being adjusted and at the same time the control pulses AEP and SECI are low, the adjust pulse for this tap is enabling the counters of the tap to increment or decrement.

The MSB of the error signal ($\text{sign } \varepsilon(n)$) and of the input or output signals (for FF or FB paths) are also available in each pair taps board. Then the LMS algorithm could be realized by using an exclusive-OR gate to control the direction of tap counters for counting up or down. In Fig. 9 this process was shown.

When the counters of tap become over or under flow, they stop to count automatically in order to avoid any jump from max. to min. or vice-versa.

The taps values are coded in two's complement coding system with 12 bits. The input or output samples contain 8 bits. In order to obtain the weighted samples of input or output in the FF or FB paths, we need a fast 12 x 8 multiplier and here we have used a pair of 8 x 8 bits multipliers for each tap. Then the results of the multipliers are added together in order to realize a weighted sample with 12 bits. In Fig. 17 the weighting process for each tap is shown.

In order to cancel a ghost with long delay time (far ghost) it is necessary to increase the tap number. Since in the LMS algorithm the gradient noise and misadjustment factor of the taps increase proportionally to the taps number [6] it is advisable to insert a flat delay of length slightly shorter than the delay of the ghost signal. In other words we need a variable delay line which is inserted in tapped delay line. The variable delay line can be realized by a pair of Random Access Memory (RAM) chips, using the 2 chips alternatively in the write mode and the read mode. The number of clocks within the alternating interval determines the amount of delay.

The capability of LMS algorithm for multi-ghosts cancellation was studied by computer simulation of the model in the previous parts. Of course, the multi-ghosts delay times should be coincident to the taps positions in the delay line for a better performance of multi-ghosts cancelling.

In Fig. 18-a a photograph of the realized digital filter for TV ghost cancelling is shown. A board with functions for 2 taps is shown in Fig. 18-b.

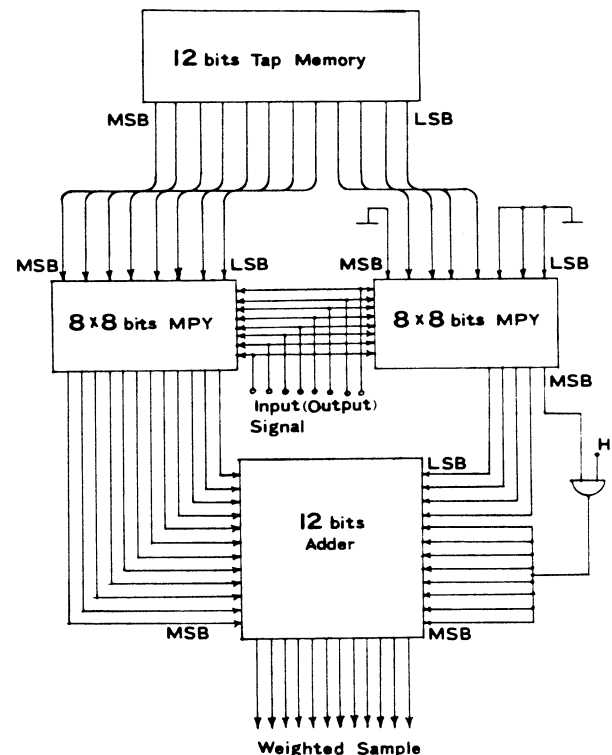


Fig. 17 Multiplication process for each tap

VI. EXPERIMENTAL RESULTS

After realization of the filter we have applied a test video ghosted signal, which is generated by ghost generator circuit^[9], for being processed. In Fig. 19-a, b a sync. pulse and color burst (test signal) before processing and after that are shown by photos of oscilloscope. The image quality can be compared in Fig. 20-a, b after cancellation a 25%, or 50% 1.6 μ sec. ghosted picture.

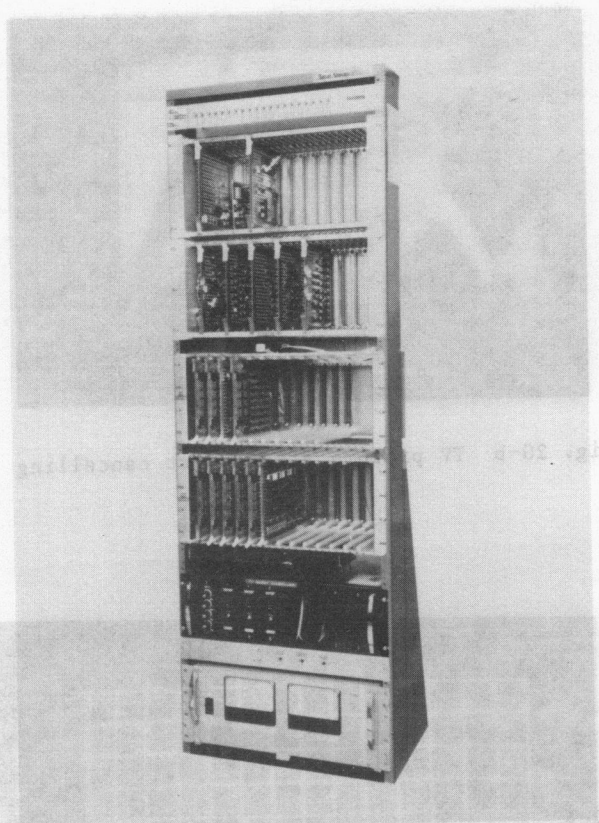


Fig. 18-a Realized LMS-RAT digital filter for TV ghost cancelling

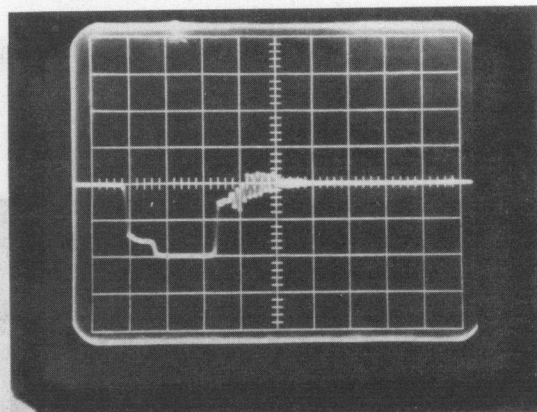


Fig. 19-a Sync. pulse & color burst (test signal) with 25% ghost at 1.6 μ sec.

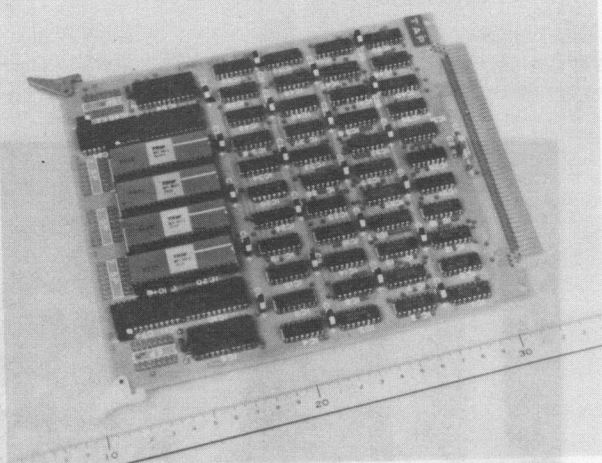


Fig. 18-b A couple of taps of the digital filter

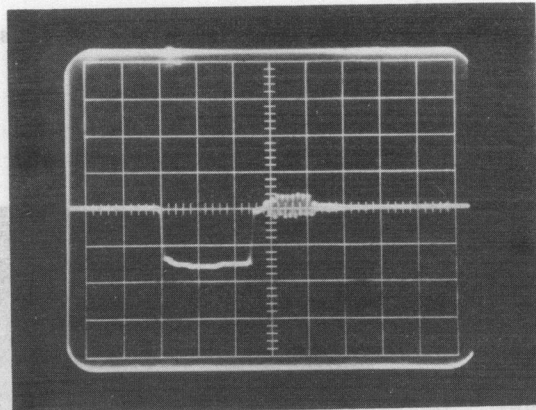


Fig. 19-b Sync. pulse & color burst (test signal) after processing in the digital filter

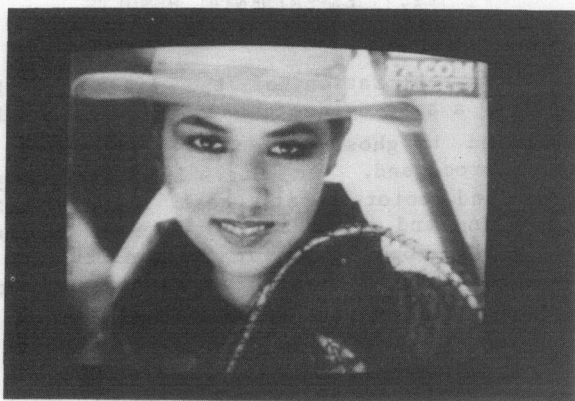


Fig. 20-a TV picture with 25% ghost at $16\mu\text{sec}$.



Fig. 20-b TV picture after ghost cancelling

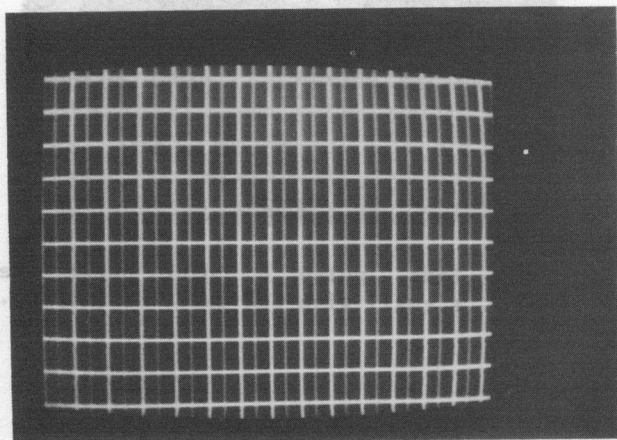


Fig. 20-a TV picture with 50% ghost at $1.6\mu\text{sec}$.

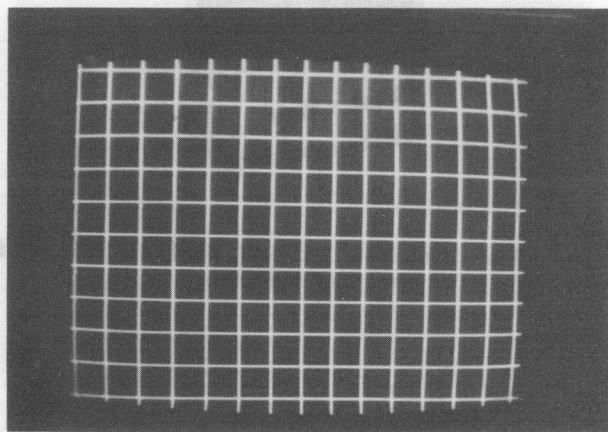


Fig. 20-b TV picture cancellation of 50% ghost

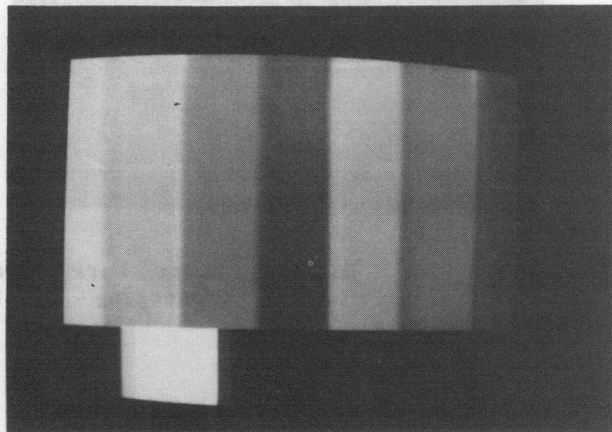


Fig. 20-a TV picture with 25% ghost at $16\mu\text{sec}$.

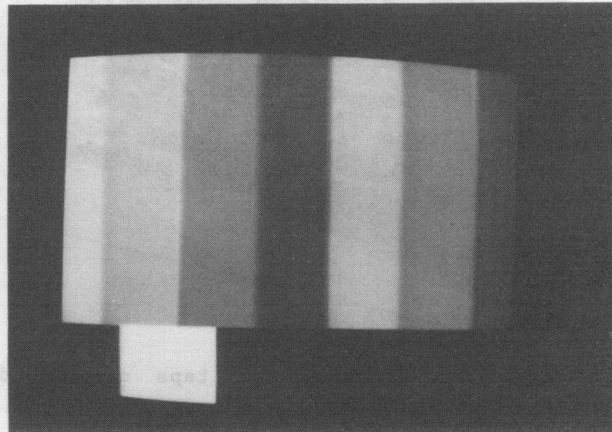


Fig. 20-b TV picture after ghost cancelling

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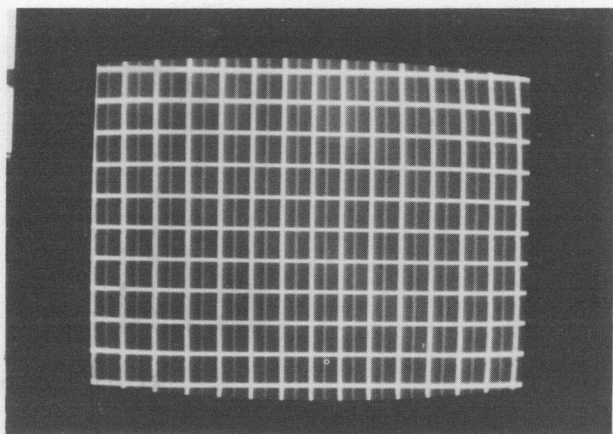


Fig. 20-a TV picture with 25% ghost at 16μsec.

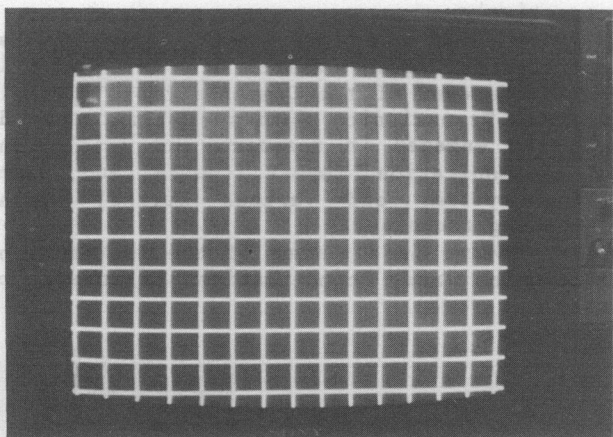


Fig. 20-b TV picture after ghost cancelling

VII. CONCLUSION

The TV ghost cancelling at the video stage by processing video signal in the adaptive recursive LMS digital filter was studied. Considering the minimization of ghost and white noise leads us to use a test signal which its power spectrum is more similar to video signal power spectrum.

The RAT control gives good performance in order to avoid the algorithm to be over-compensated or enters into a pseudo optimum condition. The stability of the RAT control is improved by a new signature operator, namely, lazy sign. The computer simulation results confirm the capability of proposed methods for controlling the algorithm.

The hardware realization of the TV ghost canceller approves the feasibility and capability of the adaptive recursive LMS digital filter with proposed control methods by visual evaluation of the processed video signal.

BIOGRAPHIES



Mohammed Reza Asharif

Mohammad Reza Asharif (S.M.) was born in Teheran, Iran, on December 15, 1951. He received the B.S., M.Sc. degrees in electrical engineering from the University of Teheran, Teheran in 1973, 1974, respectively and the Ph.D. degree in electrical engineering from the University of Tokyo, Tokyo, in 1981.

From 1975 to 1977, he joined to National Iranian Radio & TV (NIRT) broadcasting, where he was a chief engineer of maintenance and repairment of TV & FM transmitters in Tabriz and Urumie (north-west prefectures of Iran).

Since 1979, he has been a research student and later a trainee in the Fujitsu Laboratories Ltd., Kawasaki, in parallel with studying at the University of Tokyo.

Dr. Asharif is a member of the Institute of Electronics and Communication Engineers of Japan.

His research interests are in the field of digital signal processing and TV ghost cancelling.



Kazuo Murano

Kazuo Murano (S'62-M'72) was born in Nara, Japan, on November 5, 1945. He received the B.E. degree from University of Tokyo, Tokyo, Japan in 1968 and M.A. and Ph.D. degrees in Electrical Engineering from Princeton University, Princeton, N.J. in 1970 and 1972, respectively.

Since 1972 he has been with Fujitsu Laboratories Ltd., where he is currently Section Manager of the Communication Terminals Section, Transmission Systems Laboratory. He has been engaged in research and development on communication theory, voice and video coding system, and various digital signal processing technique.

Dr. Murano is a member of the Institute of Electronics and Communication Engineers of Japan.



Mitsutoshi Hatori

Mitsutoshi Hatori (R.M.) was born in Tokyo, Japan, on September 22, 1938. He received the B.S., M.S. and Ph.D. degrees in electrical engineering from the University of Tokyo, Tokyo, in 1963, 1965 and 1968, respectively. In 1968 he joined Department of Electrical Engineering, the University of Tokyo, where he is presently Associate Professor. From 1975 to 1976 he stayed at Technische Universitaet Muenchen as stipendiat of Alexander von Humboldt Stiftung. His research interests are in cancelling of TV ghost, cancelling of FM multipath effect, picture processing, picture coding and also in radiography. He is a member of the Institute of Electronics and Communication Engineers of Japan, the Institute of Television Engineers of Japan and the Institute of Electrical Engineers of Japan.